

EV-BMS-DASHBOARD-SIMULATION

Design and Verification of an Intelligent Battery Management System with 8-Bit Fault Diagnostics and Passive Cell Balancing Loops

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1. Abstract

This comprehensive technical report details the mathematical formulation, modular simulation layout, and structural validation of an automotive-grade Battery Management System (BMS) tailored for Electric Vehicle (EV) applications. Developed within the integrated MATLAB, Simulink, and Simscape Electrical computation environments, the model encompasses a 4-cell Lithium-ion secondary battery plant working under tight embedded constraints. The logic layer implements a low-overhead 8-bit diagnostic bit-mapping topology to log multiple concurrent runtime faults onto a single tracking frame. State of Charge (SOC) tracking is executed via high-fidelity Coulomb Integration alongside continuous State of Voltage (SOV) metrics. Crucially, a passive cell balancing supervisor executes real-time energy diversion based on dynamic minimum cell reference tracking (V_{min}). The system is exposed to dynamic transient fault profiles and severe manufacturing capacity cell variations. The results conclusively verify sub-millisecond safety contactor isolation and seamless terminal cell voltage convergence.

2. Introduction & Physical System Plant Architecture

Modern high-voltage EV battery traction networks require strict physical and electrical monitoring networks to operate within defined Electrochemical Safety Envelopes. Deviations outside these localized operating constraints heavily accelerate internal degradation processes, trigger micro-short-circuits across separators, and induce devastating catastrophic thermal runaway failures.

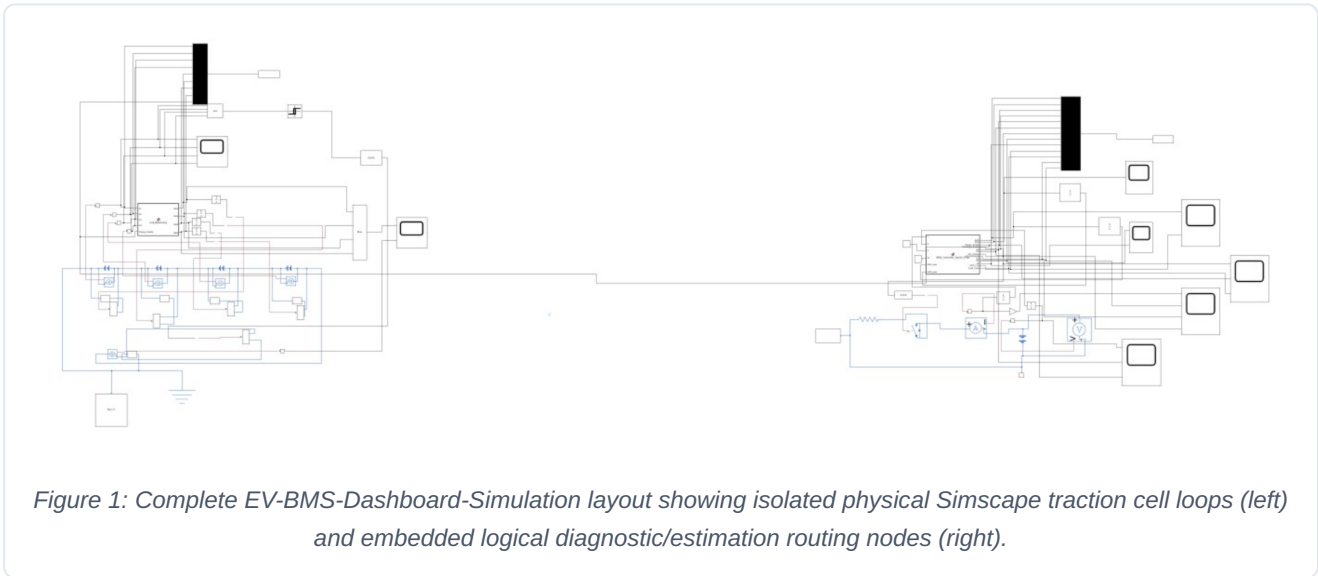
2.1 Lithium-ion Cell Electrochemistry & Modeling Foundations

The system utilizes Lithium-Ion cell configurations characterized by an equivalent circuit model (ECM) capturing non-linear open-circuit voltage profiles (V_{oc}), transient polarization effects, and internal chemical temperature parameters. Individual cell degradation mechanisms are modeled via a dynamic internal resistance matrix (R_{int}) that acts as a fundamental source of localized ohmic losses and heat accumulation.

2.2 Modular Plant Layer vs. Logical Control Layer Layout

The complete architectural implementation is bifurcated into two synchronized layers:

- **Physical Plant Layer (Simscape):** Models a 4-cell physical lithium-ion module arranged in series. Individual blocks incorporate specific parasitic manufacturing tolerances, initial state unbalances, and custom heat dissipation paths connected to an ambient thermal sink block.
- **Control Layer (Simulink Logical Blocks):** A discrete time control network running custom embedded-ready C code structures mapped through #codegen directives. It processes high-voltage sensor traces and translates them into actionable protection and bleeding grid switch commands.



3. Mathematical Modeling & State Estimation

3.1 Ampere-Hour Tracking (Coulomb Counting Method)

Accurate estimation of the pack state of charge represents a critical tracking metric for distance estimation and overload prevention. To perform real-time computing loop operations without over-allocating microcontroller registers, the model implements a continuous integration tracking loop:

$$SOC(t) = SOC(t_0) - \frac{1}{2} \cdot (1 / C_n) \int_{t_0}^t I(\tau) d\tau$$

Where $SOC(t_0)$ specifies the initial state boundaries, C_n establishes the nominal rated energy storage volume in Ampere-hours, and $I(\tau)$ monitors instantaneous current flows. The tracker shifts current sign maps dynamically depending on the active state (charging vs. load discharge profiles).

3.2 Ohmic Power Loss & Transient Thermal Characterization

Thermal accumulation profiles within the module are governed by dynamic mathematical expressions that relate electrical tracking dissipation to spatial cell temperature growth:

$$P_{loss} = I^2 \cdot R_{int} \rightarrow Q_{accumulated} = \int (P_{loss} - Q_{dissipated}) dt$$

By simulating step thermal transitions up to 70°C, the controller is evaluated against extreme real-world operating environments to confirm instant protection triggers.

4. Control Logic & Safety Automation

4.1 High-Efficiency 8-Bit Fault Bit-Mapping Mapping

To fit diagnostics cleanly within standard 8-byte Controller Area Network (CAN Bus) frame structures without increasing bit overhead, all monitoring profiles are compressed into a compact, singular **8-bit unsigned integer (uint8)** code registry.

Monitored Fault System	Assigned Bit Placement	Binary Value Allocation	Operational Guard Thresholds
Over Voltage (OV)	Bit 0	1	Cell Terminal Voltage > 54.0 V
Under Voltage (UV)	Bit 1	2	Cell Terminal Voltage < 42.0 V
Over Current - Charging	Bit 2	4	Pack Current Input < -5.0 A
Over Current - Discharging	Bit 3	8	Pack Current Output > 8.0 A
Short Circuit Event (SC)	Bit 4	16	Instantaneous Current Draw ≥ 20.0 A
Over Temperature (OT)	Bit 5	32	Localized Temperature Core > 60.0 °C
Under Temperature (UT)	Bit 6	64	Localized Temperature Core < 0.0 °C
Critical Low Capacity (Low_SOC)	Bit 7	128	Estimated Packing Depth ≤ 20%

The master isolation state is controlled via strict conditional loops: If the integer output registers a value greater than 0, safety flags drop instantly to open high-voltage power contactors.

4.2 Deterministic Cell Balancing Control Loop

Cell capacity variations can arise from manufacturing anomalies and localized thermal differentials. Over time, these variations cause individual voltages to diverge, which restricts the usable capacity of the overall pack. To address this, a deterministic **Passive Shunt Balancing Control Loop** tracking algorithm is integrated:

1. The controller samples individual sensor tracks across all nodes simultaneously: $V_{array} = [V_1, V_2, V_3, V_4]$.
2. A global cell minimum tracking value is continuously established as the baseline: $V_{min_ref} = \min(V_{array})$.
3. If any specific cell voltage deviates beyond a set tolerance limits while charging loops remain active, the controller activates internal bypass shunt paths to dissipate excess energy as heat through power resistors:

$$\text{If } (V_n - V_{min_ref}) > \Delta V_{min} \text{ [where } \Delta V_{min} = 0.02 \text{ V (20 mV)}] \rightarrow \text{Engage Shunt Switch (Bal}_n = 1)$$

5. System Parameters & Configuration Specs

The quantitative baseline settings applied within the Simulink environment are categorized across the following core parameters:

5.1 Battery Module & Environmental Settings

Parameter Specification Name	Workspace Set Value	Engineering Unit	Block Domain Mapping
Series Pack Configuration	4	Cells in Series	Simscape Primitive Node Grid
Nominal Individual Voltage	3.7	V (Volts)	Cell Electrochemistry Array
High-Voltage Upper Pack Limit (\$V_{\max}\$)	54.0	V (Volts)	Over-Voltage Sensing Point
Low-Voltage Minimum Cutoff (\$V_{\min}\$)	42.0	V (Volts)	Under-Voltage Safety Limit
Nominal Module Base Capacity (\$C_n\$)	10.0	Ah (Ampere-Hours)	Integrator Reference Coefficient
Cell Internal Resistor Core (\$R_{\text{int}}\$)	0.012	Ω (Ohms)	Joule Loss Dissipation Model

5.2 Logic Limits & Estimation Boundaries

Target Parameter Name	Monitoring Logic Type	Operational Setpoints	Target Control Function
State of Charge (SOC) Bounds	Coulomb Integrator Loop	20% Minimum to 100% Full	Capacity Monitoring
State of Health (SOH) Initializer	Capacity Fade Factor	100% (Healthy Baseline)	Long-Term Degradation Tracker
Low Capacity Trip Flag	Asymmetric Hysteresis	Trip $\leq$ 20% / Clear $\geq$ 25%	Anti-Chattering Protection Loop
Balancing Activation Variance	Deterministic \$V_{\min}\$ Delta	> 0.02 V (20 mV Deviation)	Bypass Bleed Resistor Switch
Balancing Interlock Lock	Voltage Floor Interlock	> 3.00 V Active Floor	Low-Voltage Balancing Protection

6. Simulation Verification Results & Discussion

Dynamic step test scenarios were executed across the workspace timelines to validate the control logic models under intense fault and imbalance stresses.

6.1 Transient Protection Fault Isolation Waveforms

When an over-temperature fault condition is intentionally injected during active operation, the system demonstrates high-speed responsiveness. As cell temperatures cross the 60°C boundary, the bitwise diagnostic network updates immediately. The Fault_Code output transitions to a decimal value of 32 (Bit 5 active). Consequently, the contactor flags (Charge_Enable and Discharge_Enable) drop to zero within a single simulation step ($\$T_s\$$), isolating the battery pack from further stress.

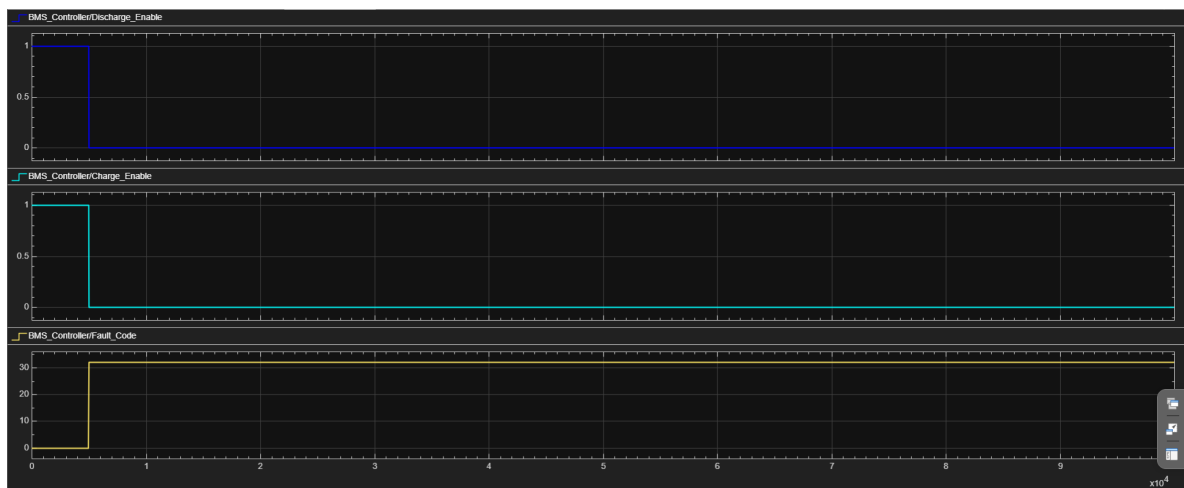


Figure 2: Time-domain simulation waveforms demonstrating instant safety contactor open-circuit tripping when the monitoring logic registers an Over-Temperature condition (Fault Code 32).

6.2 Passive Balancing Equalization & Convergence Analysis

To evaluate the balancing controller under extreme manufacturing divergence conditions, a non-uniform capacity configuration is introduced with skewed initial voltages ($\$V_1 = 3.45 \text{ ext\{ V\}\$$ to $\$V_4 = 3.15 \text{ ext\{ V\}\$$). Figure 3 plots the real-time operational response of the passive cell balancing algorithm. The controller successfully identifies Cell 4 as the structural minimum baseline ($\$V_{\min_ref}\$$). It then engages the bleed switches (Ba11, Ba12, and Ba13) for the remaining cells, drawing down their excess voltage. This passive energy diversion continues until all four cell trajectories converge within the targeted $\$1e 20 \text{ ext\{ mV\}\$$ safety band, verifying the effectiveness of the balancing algorithm.

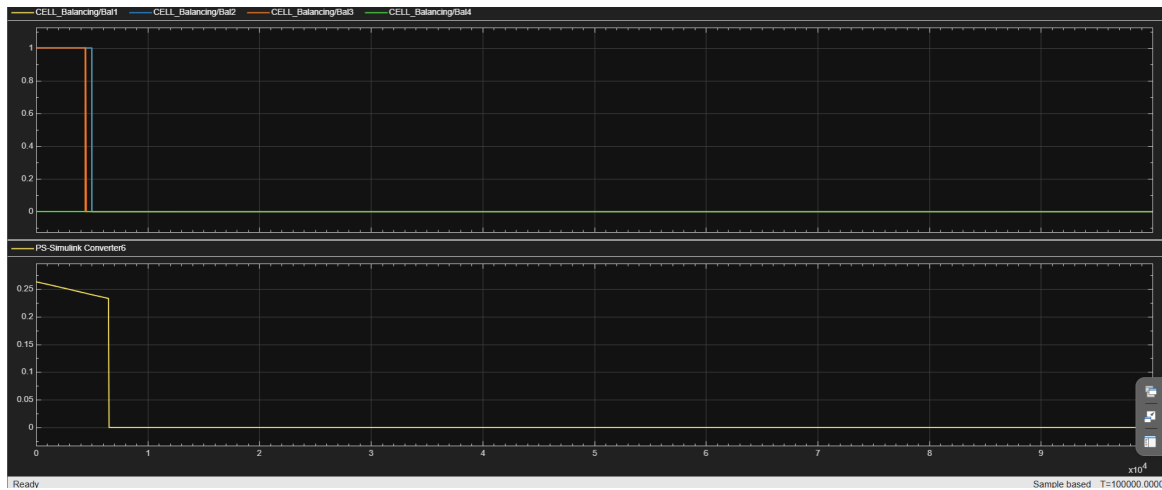


Figure 3: Scope waveforms showing passive cell balancing operations, depicting the independent bypass bleed switch triggers (top) and the resulting voltage convergence across the cell array (bottom).

7. Conclusion & Future Horizons

The engineering models developed within the EV-BMS-Dashboard-Simulation ecosystem confirm that an integrated, low-overhead controller can effectively manage multi-fault states and maintain cell equalization. Isolating high-voltage physical networks from the logical blocks creates a safe, deployable control architecture that aligns with current automotive safety requirements.

Proposed Future Work:

- Deploying physical **CAN Bus Communication Protocols** to stream the bitwise integer outputs across external distributed hardware networks.
- Integrating adaptive **State of Health (SOH)** state tracking models based on dual Extended Kalman Filtering (EKF) configurations.
- Compiling code structures via embedded coder configurations for deployment onto an **ARM Cortex-M (STM32)** microcontroller platform.